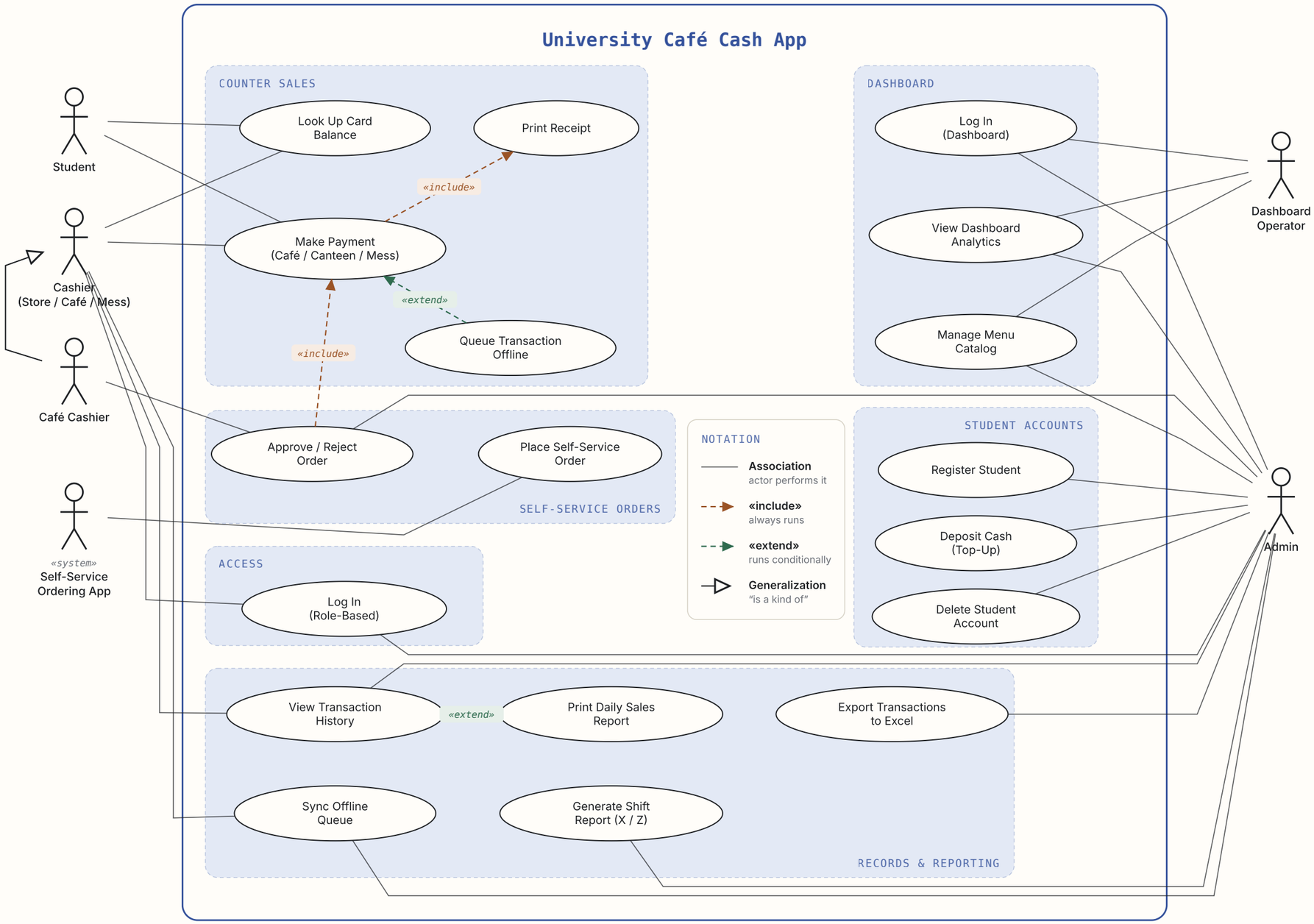
TRANSACTIONSYSTEM · UML

Use Case Diagram

Six actors against eighteen use cases, verified line by line against the live WinForms codebase — the role→tab wiring in **Form1.cs**, and the actual features inside each control.



Grouped by subsystem. Every association mirrors a tab the role actually gets in `Form1.InitializeDynamicUI`; every «include» and «extend» mirrors a call the code actually makes.

Why it is drawn this way

- **Admin is not a cashier.** `Form1` gives the Admin role the Dashboard, Cafe Orders, Admin and Transactions tabs — but *not* the Student Store tab. So Admin never appears on Look Up Card Balance or Make Payment, and does appear on Approve / Reject Order and the whole Dashboard group.
- **Café Cashier is a specialization, not a separate role.** Store, Café and Mess all log in the same way and get the same Student Store and Transactions tabs; only `role == "Cafe"` additionally gets the Cafe Orders tab, so Café Cashier inherits from Cashier and adds one use case.
- **Two independent logins.** Cashier and Admin authenticate through `LoginForm`'s role check; the Dashboard has its own hardcoded credentials inside `DashboardControl`, which even Admin must pass to open the Dashboard tab.
- **Approve / Reject Order includes Make Payment.** Approving a self-service order calls `MakePaymentByDocIdAsync` — the same charging path as a counter sale, not a separate mechanism.
- **Offline is an extension, not a separate flow.** `MakePaymentOfflineSafeAsync` only queues to `offline_queue.json` when the network check fails, which is exactly what «extend» means. Sync Offline Queue is its own use case because the status bar's "Sync Now" button is available to every role.
- **The card is the master record.** Look Up Card Balance, Make Payment, Register Student, Deposit Cash and Delete Student Account all read and write the MIFARE card directly (`ReadStudentData` / `WriteStudentData`); the balance shown at the counter comes from the card, not the database.
- **Place Self-Service Order belongs to an external system.** Nothing in this codebase inserts into the `orders` table — a separate client does, and this app only approves against it.